

Lecture 01

Introduction to Recommender Systems

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Today's Class

- Introduction to Recommender Systems
 - Motivation for Recommender Systems
 - Feedback Types: Explicit & Implicit Feedback
 - Concept of Collaborative Filtering



Motivation for Recommender Systems

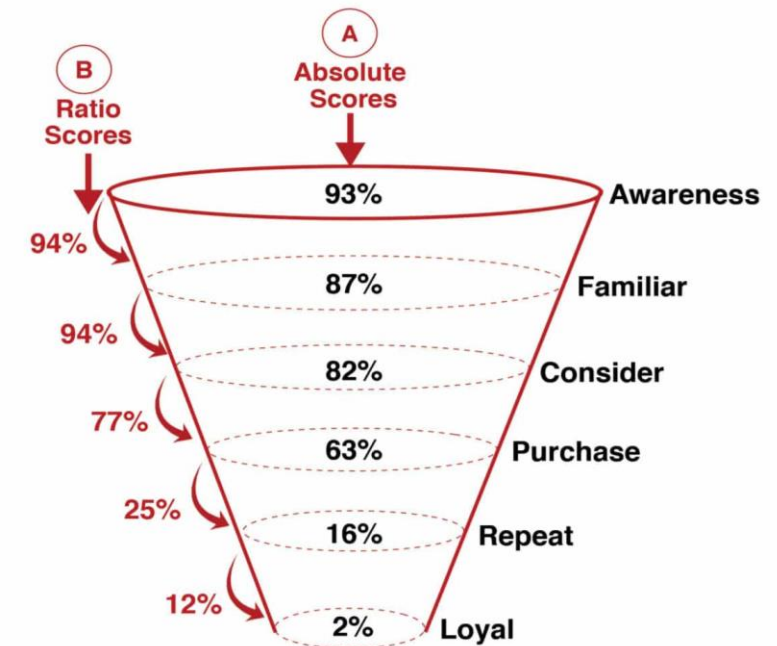
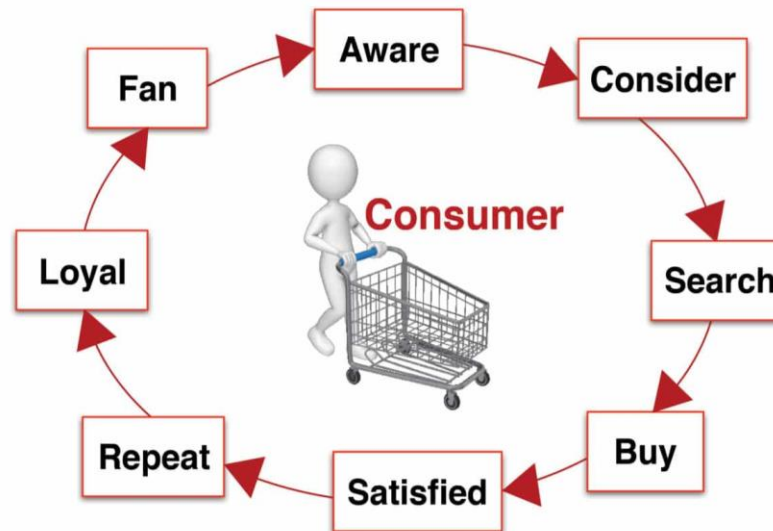
Motivation for Recommender Systems

- User Behaviors on Online Services
 - Current online services have a large number of items up to million-scale (e.g., Amazon has 12 million items and Netflix has 17,000 titles, ...)
 - However, users have limited resources (e.g., time, budget, etc.) to spend on items
 - Ironically, many users spend a lot of time carefully choosing items when there are many options (items) available

Motivation for Recommender Systems

- Funnel Analysis

- Data analysis technique that tracks conversion rates at each step of the user journey to understand where users drop off
- Users are more likely to churn as the process to achieve satisfaction with the service becomes longer



Motivation for Recommender Systems

- Aha Moment

- The moment when a user experiences the core value of a service
- Users who experience an 'Aha Moment' are more likely to use the service again



Motivation for Recommender Systems

- Recommending relevant items to users helps provide a satisfying experience
 - Recommending items that users will like among countless options is crucial for reducing users' decision-making time and enhancing satisfaction
 - For this purpose, it is important to **understand user preferences** and accurately **estimate user preferences** for items

→ **Increasing revenue and user retention!**



Feedback Types: Explicit & Implicit Feedback

Feedback Types

- Recap: Understanding User Preferences
 - The essence of recommendation systems lies in understanding user preferences
 - The basic principle of recommendation is that **user preferences can be captured from user action data on items**



Feedback Types

- **User Feedback on Items**

- If users provide feedback on items (e.g., likes or dislikes), it becomes easier to understand their preferences for those items

Feedback Types

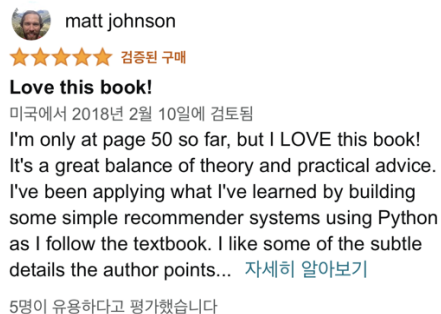
- **Explicit Feedback**

- Explicit feedback refers to **direct evaluations or ratings** provided by users, such as reviews and star ratings

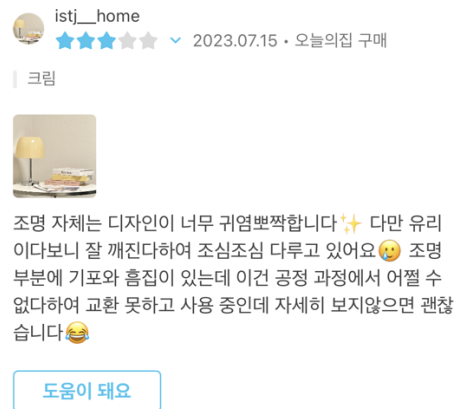
Feedback Types

- **Explicit Feedback Examples**

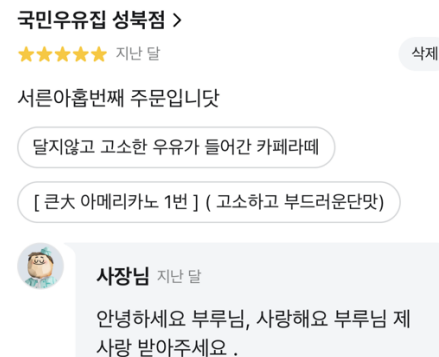
- **Rating system** is a method used to quantify user preferences and opinions, commonly used in recommender systems and applied across various fields
- Users rate/evaluate products or services based on their experiences, indicating the quality or level of satisfaction
 - In general, **lower ratings represent dissatisfaction**, while **higher ratings indicate satisfaction**



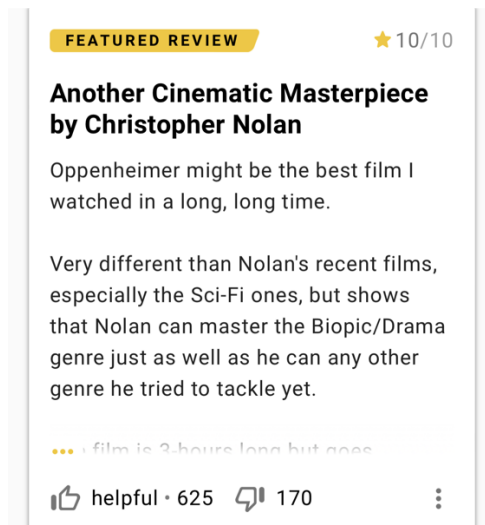
<Book Review>



<Product Review>



<Restaurant Review>



<Movie Review>

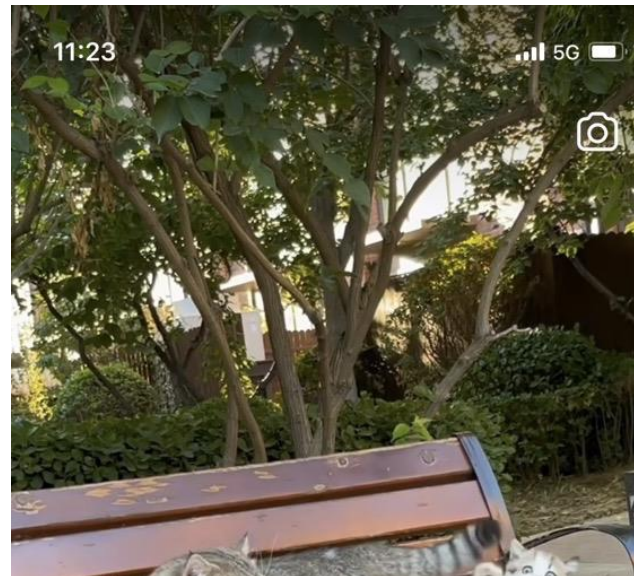
Feedback Types

• Explicit Feedback Examples

- **Like system** is a method where users click a 'Like'/'Dislike' button or symbol to express a positive/negative reaction to a specific item or piece of content
- It is a method for expressing user preferences or likability through simple symbols or indicators



<Linkedin Post>



<Reels Video>



<YouTube Video>

Feedback Types

- **Explicit Feedback: Pros**

- **Clear Opinion Expression**

- Users can explicitly indicate their preferences and tastes by providing direct ratings or writing reviews

- **Enhanced User Understanding**

- Explicit feedback helps in understanding user preferences

- **Training and Test Data**

- Explicit feedback can be used to train and evaluate the performance of recommender systems

→ **more accurate and personalized recommendations!**

Feedback Types

- **Explicit Feedback: Cons**

- **Sparse Data**

- Many users may not provide explicit ratings or reviews for products

- **Lack of Consistency**

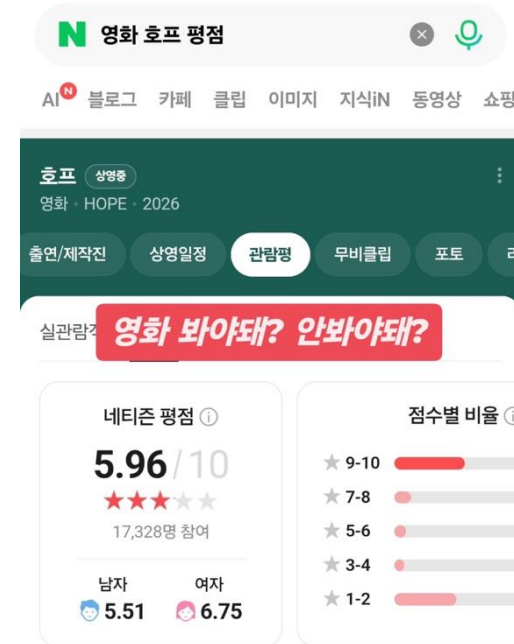
- Users may assign different ratings to the same item at different times

- **Subjectivity of Ratings**

- User ratings and reviews can be subjective

- **Difficulty in Handling New Items**

- The lack of explicit feedback for new items



Feedback Types

- **Implicit Feedback**

- Users' **implicit actions and patterns** on specific items, even if they haven't directly evaluated them

Feedback Types

- **Implicit Feedback Examples**

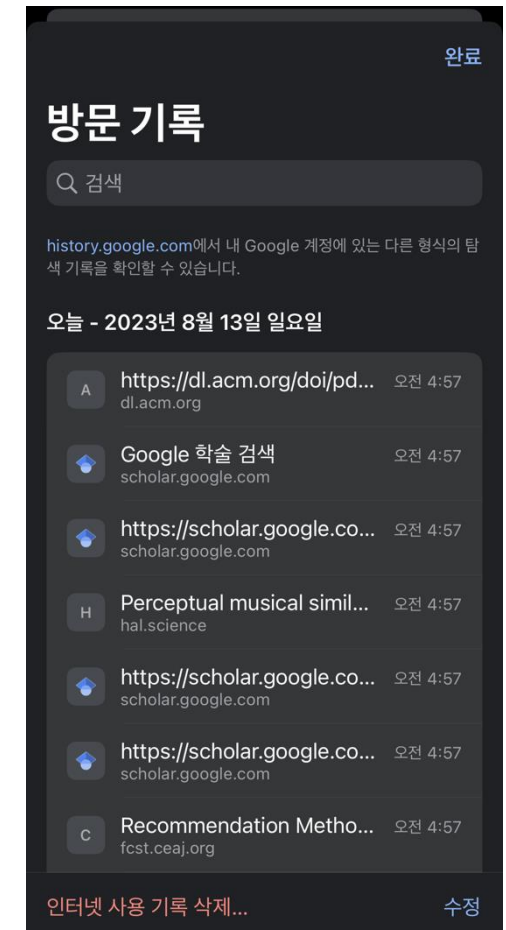
- **Purchase History:** Understanding user preferences based on the list of products or services they have previously purchased



Feedback Types

- **Implicit Feedback Examples**

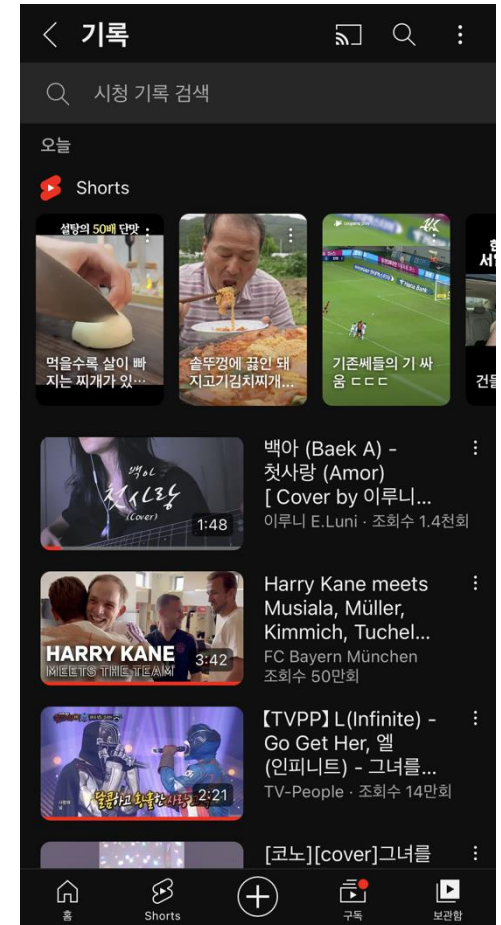
- **Page Views and Clicks:** Inferring user interests by analyzing the items they have viewed or clicked on web pages



Feedback Types

- **Implicit Feedback Examples**

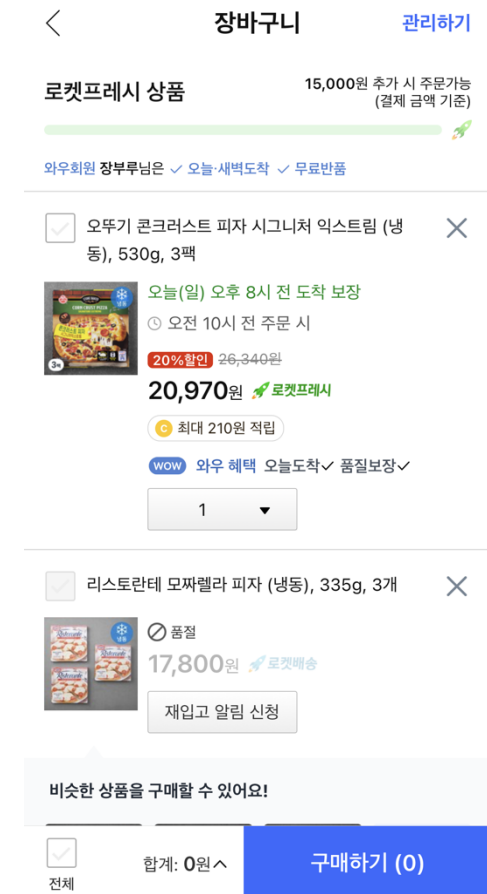
- **Viewing History:** Inferring user preferences by analyzing the videos they have watched on video streaming platforms



Feedback Types

- **Implicit Feedback Examples**

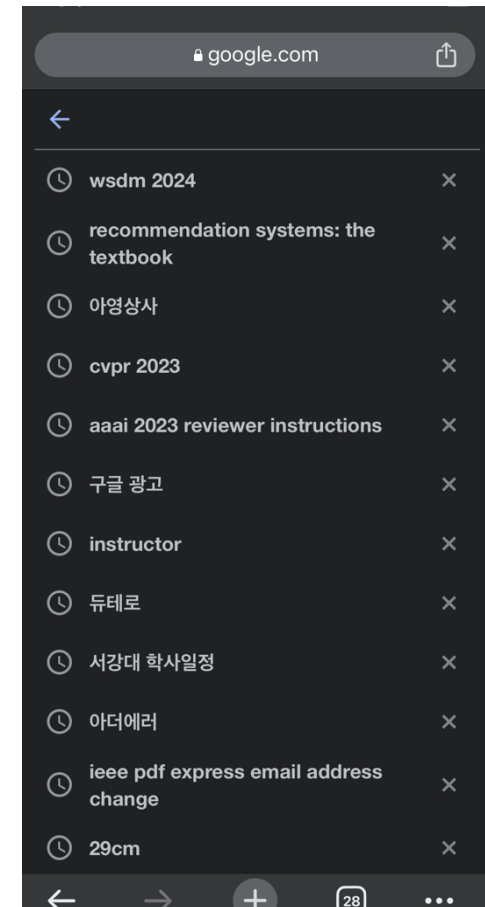
- **Adding to Cart:** Gauging user interests based on the items added to their online shopping cart



Feedback Types

- **Implicit Feedback Examples**

- **Search History:** Inferring user interests from the keywords or queries they have searched



Feedback Types

- **Implicit Feedback: Pros**

- **Ease of Data Collection**

- Implicit feedback is easier to collect than explicit feedback because it leverages users' everyday actions and patterns. This advantage can increase the volume of feedback

- **Sparsity Mitigation**

- Explicit feedback often suffers from sparsity because users do not always provide ratings intentionally. In contrast, implicit feedback leverages user interactions, which helps alleviate sparsity issues

- **Natural User Experience**

- Users' interests and preferences can be reflected even without explicitly rating items, leading to a more seamless and natural experience with recommender systems

- **Exploring Diverse Items**

- By analyzing users' diverse interactions, recommender systems can discover new items and provide users with a variety of experiences

Feedback Types

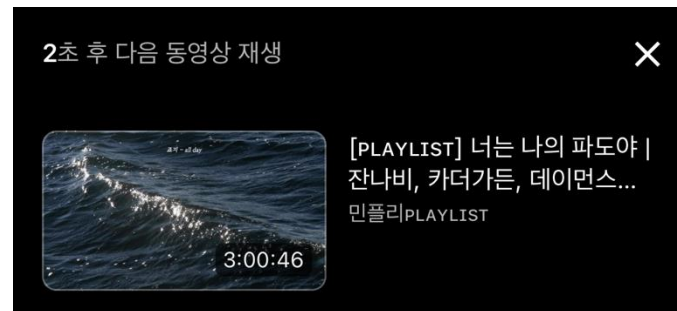
- **Implicit Feedback: Cons**

- **Difficulty in Interpretation**

- Interpreting implicit feedback can be challenging, as it is based on user actions without explicit intent

- **User Intent Ambiguity**

- Implicit user actions may not always accurately reflect their true intentions. For example, clicking on a page does not necessarily indicate a genuine interest in the item



- **Challenges in Handling Diverse Environments**

- Since implicit user actions occur in various contexts, processing them effectively may require complex algorithms and technologies

Feedback Types

- **Combining Explicit & Implicit Feedbacks**

- **Ensuring Information Completeness**

- Explicit feedback reflects users' direct opinions, while implicit feedback is derived from analyzing user behavior. Combining both types of feedback can enhance the completeness of information

- **Overcoming Sparsity Issues**

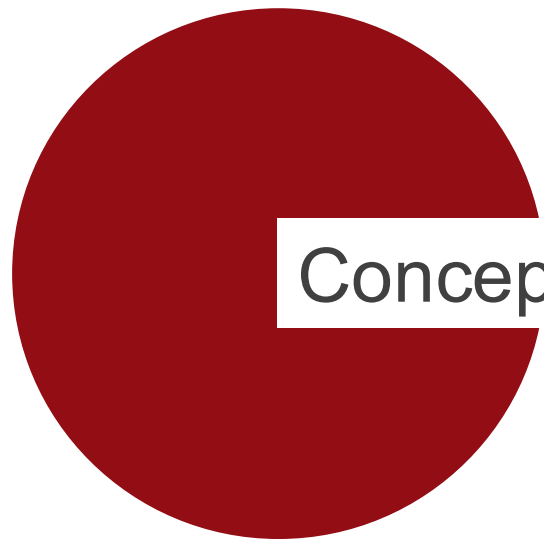
- While explicit feedback can partially alleviate data sparsity, implicit feedback helps collect additional data, thereby improving the performance of recommendation algorithms

- **Considering Various Aspects**

- Explicit feedback provides insights into specific aspects related to user evaluations of items, while implicit feedback, derived from users' everyday behavior, offers a broader perspective

- **Enhancing Reliability**

- By comparing and combining explicit and implicit feedback, the reliability of recommender systems results can be improved



Concept of Collaborative Filtering

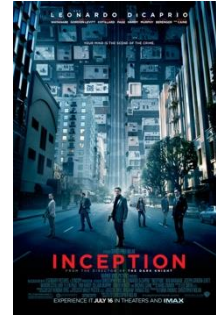
Concept of Collaborative Filtering

- **Motivation for Collaborative Filtering**

- The motivation for collaborative filtering stems from the idea that people often receive the best recommendations **from others with similar tastes**
- Collaborative filtering encompasses techniques for **matching individuals with similar interests** and making recommendations based on this information

Concept of Collaborative Filtering

- Example

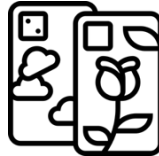


	5	4		2	2		
		2	3	5		4	
	5	?	5	3	2	?	

<Movie Rating (Explicit Feedback)>

Concept of Collaborative Filtering

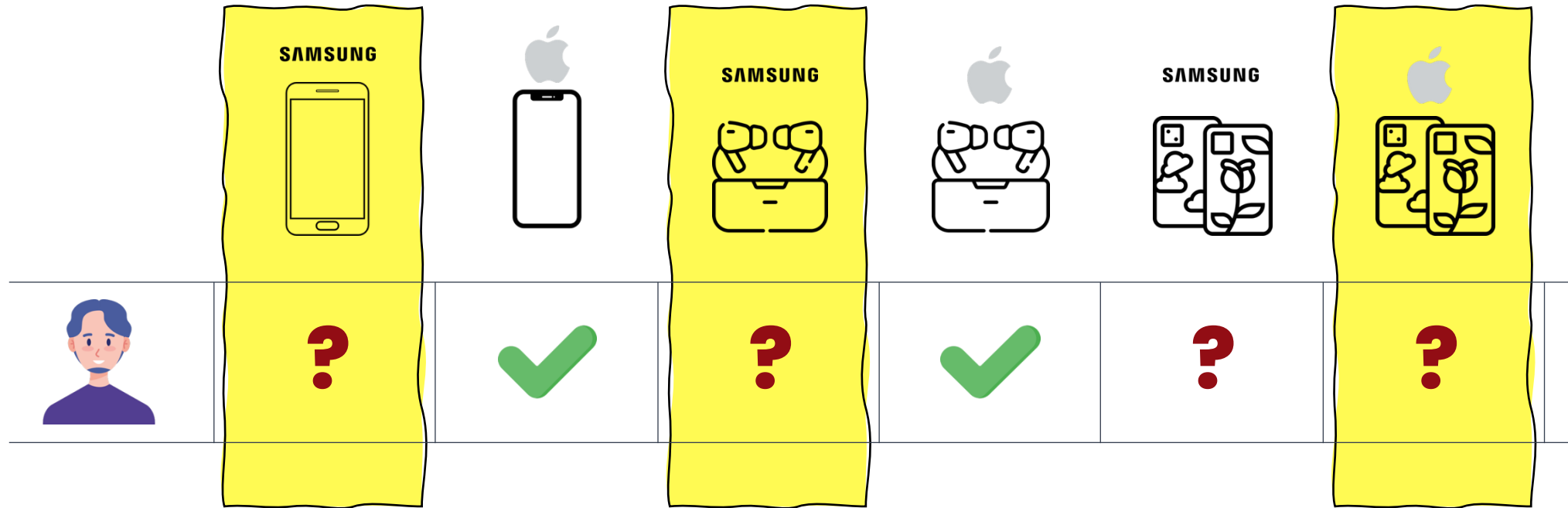
- Example

	SAMSUNG 		SAMSUNG 		SAMSUNG 	
	✓				✓	
		✓				✓
	?	✓	?	✓	?	?

<View History(Implicit Feedback)>

Concept of Collaborative Filtering

- Example



Concept of Collaborative Filtering

- **The Essence of Collaborative Filtering**

- Identifying users with similar preferences or finding similar items
- To achieve this, **measuring the similarity** between users or items is essential
- To model user or item similarity, it is crucial to **design effective representations for both users and items**

Concept of Collaborative Filtering

- Two Main Methods of Collaborating Filtering
 - **Memory-based Methods:** The earliest collaborative filtering algorithms that predict the ratings of user-item combinations **based on their neighborhoods**. These methods do not require any model training.
 - **Model-based Methods:** Machine learning and data mining techniques are used to develop predictive models. These **models are trained to learn user preferences based on user action data**.

Concept of Collaborative Filtering

- In this course,

